

Presentation and assessment of the project work

The survey, the pie chart and the conclusions — presented, questioned and marked.

LESSON 203–204

2 × 40 MIN

MATEMATIKA 6, LOYIHA HIMOYASI

S7 PROJECT PRESENTATION

LESSON CLOCK

10'

25'

25'

20'

What is presented	10 min
The presentations	25 min
The assessment criteria	25 min
Questions, and closing the year	20 min

LEARNING OBJECTIVES

- Present a survey, its chart and its conclusions in four minutes.
- Justify the sample and state one honest limit of the result.
- Answer questions about the data with the numbers in front of you.
- Assess another project against the published criteria.

TEXTBOOK

National	pp. 581–584
Cambridge	Stage 7, pp. 184–188
Workbook	Project assessment

01

Explanation

What is presented

Each pupil brings one poster and speaks for four minutes. The mathematics of the project is everything from lessons 157 to 168, applied to data they collected themselves.

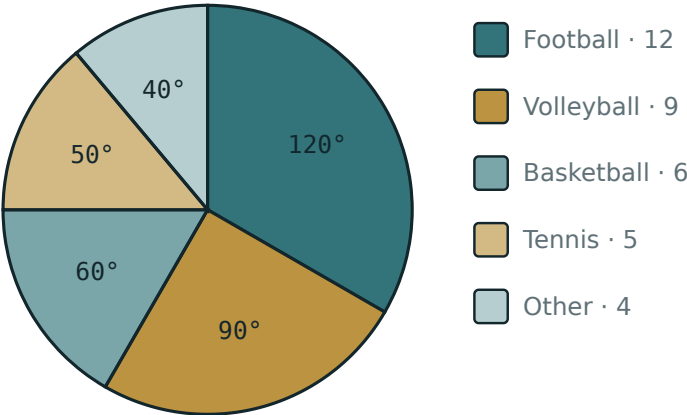
ON THE POSTER	FROM
the question, with its list of answers	L158–159
the sample: who was asked, how many, and how they were chosen	L158–159
the frequency table, with the total	L157
the pie chart, with every sector angle written on	L160–163
three conclusions and one honest limit	L164–168

THE TOTAL IS NOT DECORATION

Without the number surveyed written on the poster, no sector can be turned back into people. It is the first thing the audience will ask for.

The presentations

A worked example, from a survey of 60 pupils.



SPORT	PUPILS	ANGLE	PERCENTAGE
football	20	120°	33.3%
volleyball	15	90°	25%
basketball	10	60°	16.7%
tennis	9	54°	15%
swimming	6	36°	10%
total	60	360°	100%

ONE PUPIL IS 6°

360 ÷ 60. Every angle in the table is the frequency times six, and the check is that the column adds to 360°.

The assessment criteria

CRITERION	WHAT EARNS THE MARKS	MARKS
the question and the sample	fixed options, no gaps or overlaps; the sample justified	4
the data and the table	at least 60 asked; frequencies adding to the total	4
the pie chart	angles correct and adding to 360° ; labels or a key; the total shown	5
the conclusions	three statements from the data, and one honest limit	4
the talk	four minutes, clear, and questions answered	3
total	—	20

A BEAUTIFUL CHART WITH THE WRONG ANGLES SCORES BADLY

Five of the twenty marks are for the chart, and most of those are for the arithmetic behind it. Check the angle column against 360° before the poster is glued.

Questions, and closing the year

QUESTION THE AUDIENCE MAY ASK	WHAT A GOOD ANSWER USES
how many people is that sector?	the angle and the total
what percentage is it?	the angle divided by 3.6
would the whole school agree?	the sample, and its honest limit
if the school has 600 pupils, how many play football?	the share, scaled up: 200
why a pie chart and not a bar chart?	the data is shares of one whole

THE YEAR, IN ONE LINE

Directed numbers and fractions; expressions and equations; ratio, proportion and percentage; the circle and the solids; speed; and data with probability. Every one of them appeared in somebody's project.

02 Worked examples

EXAMPLE 1 A survey of 60 pupils gives 20, 15, 10, 9 and 6. Find the five sector angles.

- ① $360 \div 60 = 6^\circ$ for each pupil.
- ② Multiply each frequency by 6.
- ③ $120^\circ, 90^\circ, 60^\circ, 54^\circ$ and 36° .
They total 360° ✓

ANSWER

$120^\circ, 90^\circ, 60^\circ, 54^\circ$ and 36°

EXAMPLE 2 The school has 600 pupils. Estimate how many prefer football, and say what the estimate assumes.

- ① Football is $\frac{20}{60} = \frac{1}{3}$ of the sample.
- ② $\frac{1}{3} \cdot 600 = 200$ pupils.
- ③ It assumes the sample is representative of the school.
That is the honest limit ✓

ANSWER

About 200, if the sample represents the school

EXAMPLE
3**Two more pupils are asked and both choose football. Find the new sector angle, to 1 decimal place.**

① Now 22 of 62.

② $\frac{22}{62} \cdot 360.$

③ $= 127.7^\circ.$

Every other angle changes too — the total is still 360° .**ANSWER** 127.7° **03 Interactive model**

Give each pupil the criteria table and one other project to mark; explaining a lost mark to a classmate teaches the criterion better than losing it yourself.

Reading a classmate's project

With 60 surveyed, one pupil is worth:

20 of the 60 gives a sector of:

That is what percentage?

A 54° sector of 60 pupils is:

6

9

15

54

The angle column must total:

100

180°

360°

the number surveyed

Scaling 1 3 up to a school of 600 gives:

60

180

200

300

A poster with no total:

is complete

cannot be turned into numbers

is more accurate

scores full marks

Most of the chart marks are for:

the colours

the arithmetic behind the angles

the size

the handwriting

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Presentation	Taqdimot	Презентация
Poster	Plakat	Плакат
Conclusion	Xulosa	Вывод
Criterion	Mezon	Критерий
Peer assessment	O'zaro baholash	Взаимооценивание
To justify	Asoslamog	Обосновать
Limitation	Cheklov	Ограничение
Scaling up	Kengaytirib baholash	Перенос на всю совокупность

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

The first thing an audience asks for is:

the colours

the number surveyed

the title

the date

Three conclusions should come from:

opinion

the data

the textbook

the teacher

An honest limit is:

a weakness to hide

part of a good report

a lost mark

optional

A pie chart suits this project because the data is:

ordered in time

shares of one whole

two measurements

continuous

Scaling a share up to the whole school assumes:

nothing

the sample is representative

the school is small

the chart is neat

Peer assessment is done against:

a feeling

the published criteria

the largest poster

the loudest talk

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. The total surveyed in the example

ANSWER 60

2. The angle representing one pupil

ANSWER 6°

3. The sector for football, 20 pupils

ANSWER 120°

4. The sector for volleyball, 15 pupils

ANSWER 90°

5. The sector for basketball, 10 pupils

ANSWER 60°

6. The angles of the chart total

ANSWER 360°

7. The largest group in the example

ANSWER Football

Medium · the standard question

7 PROBLEMS

1. The sector for tennis, 9 pupils

ANSWER 54°

2. The sector for swimming, 6 pupils

ANSWER 36°

3. Football as a percentage

ANSWER 33.3%

4. Volleyball as a percentage

ANSWER 25%

5. Swimming as a percentage

ANSWER 10%

6. How many chose a ball game

ANSWER 45 pupils

7. ...as a sector angle

ANSWER 270°

Hard · stretch and reason

7 PROBLEMS

1. Basketball as a percentage, to 1 d.p.

ANSWER 16.7%

2. If the school of 600 followed this pattern, the footballers

ANSWER 200

3. One honest limit of this survey

ANSWER Only 60 of 600 pupils were asked

4. Why must the angle column be checked against 360° ?

ANSWER Rounding can leave the chart open

5. Two more pupils choose football: the new sector, to 1 d.p.

ANSWER 127.7°

6. The better chart for comparing the sizes of six grades

ANSWER A bar chart

7. The three things every chart must carry

ANSWER A title, labels or a key, and the total

07 Homework

Homework — the reflection

Half a page, handed in with the poster; it is read, not marked out of twenty.

1. Mark your own project against the five criteria and give yourself a total out of 20.
2. Write the one criterion you would most like to improve, and how.
3. Write the question from the audience that was hardest to answer, and your answer to it now.
4. Name the topic of the year you found most useful outside mathematics, and why.
5. Write one question about Grade 7 mathematics that you would like answered in September.